00 — Squadron Colby: Team and Roles

Mission: 02 — FST Dialogue Systems, Lovelace's Objection & Noisy Channels
Course: Conversational AI: History, Applications, Future · WiSe 2026/27 · BTU Cottbus-Senftenberg
Repository: https://github.com/xxrafidxx/Squadron-Colby---Conversational-AI

Team

Name	Squadron station	Mission 02 role	Deliverable
Tahjib Al Sakin	Data & Memory Engineer	Part 1 · Data & Payload Specialist (Historical Archivist)	`01_Lovelace_Inquiry_Brief.md`
Rafid Mohammad Ahsan	Protocol & Interface Specialist	Part 2 · Communications & Protocol Officer	`02_FST_and_Noisy_Protocol.md`
Kunal Choudhary	Security & Ethical Navigator	Part 3 · Security & Ethics Sentinel (Ethical Navigator)	`03_Entropy_Hazard_Audit.md`
Rafid Mohammad Ahsan	Core & Lead Architect	Part 4 · Systems Architect (Core Architect)	`04_fst_noisy_simulator.py`, `ps02_solution.ipynb`
Mubina Rahaman Jerin	AI Security Officer	EU AI Act compliance	`06_AI_Act_Compliance.md`

Who did what

Name	Work done
Tahjib Al Sakin	Examined Lovelace's objection (Note G) and Turing's replies (Lady Lovelace's Objection, Argument from Consciousness); designed the 5-question Anti-Turing protocol (meta-cognition, counterfactual logic, physical grounding, original insight, constrained creation) with a 0/1/2 scoring rubric; evaluated Gemini's verbatim answers against Lovelace's objection.
Rafid Mohammad Ahsan	*Protocol:* specified `FSTDialogueManager` as a Mealy transducer (INIT, GREETING, COLLECT_SLOTS, CONFIRMATION, TERMINAL/FALLBACK) with `slot_memory` and regex extraction for destination, speed and clearance; defined the noisy channel protocol (7-word limit, mandatory noise-token substitution). *Core:* built the simulator connecting the FST, noisy channel, Anti-Turing probe and 10-turn noisy interrogation to Gemini/ChatGPT/Claude, with state traces, entropy metrics, logs, tests and repository layout.
Kunal Choudhary	Ran 5 non-canonical stress tests (wrong order, sarcasm, off-topic, prompt injection, negation) against the starter FST and ours; analysed Shannon entropy, mutual information and channel capacity under noise (slot survival, 40,000 simulated dialogues); built the hazard register; ran the live Gemini run (transcript and run log).
Mubina Rahaman Jerin	Checked the simulator against the EU AI Act (whether the rule-based FST counts as an AI system, prohibited practices, high-risk classification, Article 50 transparency); reviewed API-key handling, personal data, prompt injection and unsafe actions from misunderstood input; listed fixes before deployment.

Moodle submission (3 files): `00_Team_Roles.pdf` · `Mission02_Role_Deliverables_01-03.pdf` (Parts 1–3, one section per role) · `06_AI_Act_Compliance.pdf`. Each role document is also a separate PDF in the repository (`mission-02/pdf/`). Part 4 (code, transcript, tests) is in the repository; the website FST simulator test is in `03_Entropy_Hazard_Audit.md` §2.3.